Group Topic: **Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data**

Week 4

Git Repo/ Colab Link: https://github.com/PotatoKevz/Big-Data-Analysis-Repository

Date: May 15, 2026

Objectives :

The main goal of this phase is to enhance the flood risk prediction by:

- Developing a modular machine learning pipeline that integrates with external tools for better functionality and usability.
- Creating professional and interactive visualizations of the processed data and model results.
- Presenting clear algorithm performance metrics and visualizations to demonstrate project progress.

Introduction :

This laboratory activity serves to further enhance the practical skills of the students pertaining to data handling through developing a modular machine learning pipeline that is integrated with external tools, further enhancing the prior skills gained through recent laboratory practices of handling big data. This activity implements a Flood Risk Prediction System that analyzes SYNOP weather station data from multiple stations surrounding Albay. The system utilizes machine learning models to analyze rainfall patterns and predict the probability of flood risk events. The system integrates several technologies including Python-based data processing libraries, machine learning algorithms, and an interactive visualization dashboard. The result is a comprehensive platform that transforms raw weather data into meaningful insights that can support disaster preparedness and decision-making.

Methodology :

The activity catered to a data-driven classification approach to process data using historical SYNOP meteorological data, a basis for the project objective, which is to predict rainfall-based flood risk. The methodology was executed through a structured, end-to-end pipeline, and has also implemented changes from the previous activity's methods. Due to these, some approaches from this activity were slightly modified from the earlier task documentations. This week's activity involved primary steps that were done to accomplish; developing a pipeline that utilizes external tools, and visualizing processed data in a professional format.

1. Pipeline Development with External Tool Integration

- A core backend pipeline (flood_risk_pipeline.py) was developed to handle data loading, preprocessing, feature engineering, model training, and prediction generation.
- The pipeline outputs (predictions, metrics, and model artifacts) were designed to be easily consumed by external applications.
- Streamlit was integrated as the primary external tool to build an interactive web-based dashboard (dashboard.py).
- Additional external libraries integrated include:
    - **Plotly** – for creating interactive and dynamic visualizations.
    - **Joblib** – for saving and loading the trained machine learning model (rf_flood_model.pkl).
    - **Pandas** – for efficient data loading and manipulation between pipeline and dashboard.
- This integration established a complete workflow:

  SYNOP Data → Pipeline Processing → Model Prediction → Streamlit Dashboard.

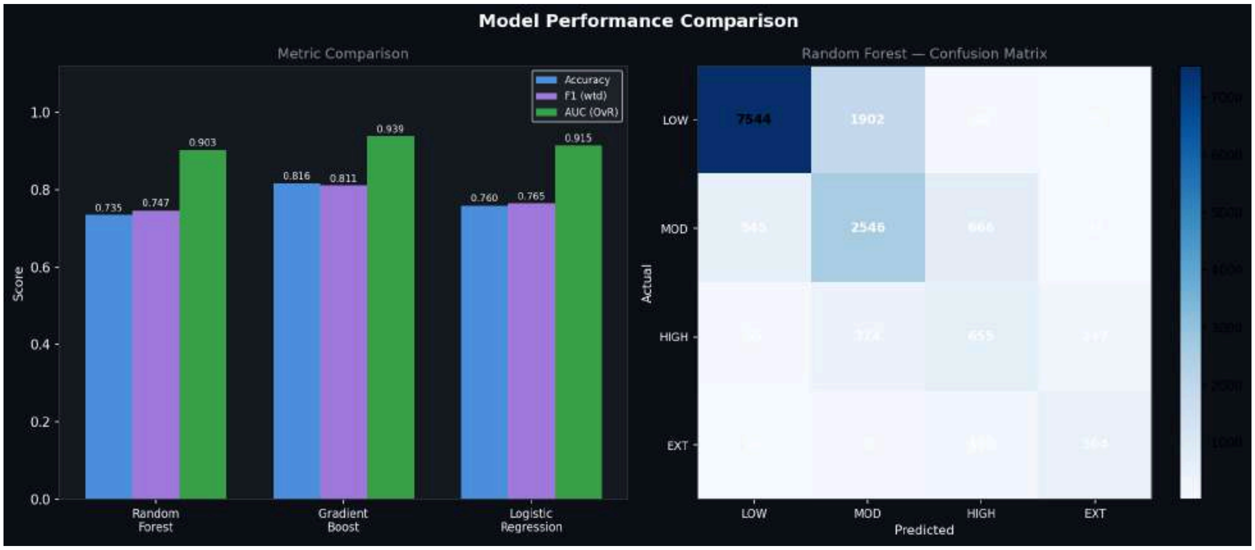
2. Professional Data Visualization

- All visualizations were generated using **Plotly** within the Streamlit dashboard to ensure high interactivity and professional quality.
- A clean, multi-tab dashboard layout was implemented with three main sections:
    - **Overview Tab**: Time series plots, risk class pie charts, rainfall histograms, and KPI cards.
    - **Model Performance Tab**: Comparative bar charts for algorithm metrics and feature importance visualizations.
    - **Extreme Events Tab**: Scatter plots highlighting high-risk and extreme rainfall occurrences.
- Interactive features such as date range filters, risk class selectors, hover tooltips, zoom/pan capabilities, and conditional formatting were added to enhance usability.
- A professional dark theme with consistent, accessible colors was applied throughout the dashboard.

Results and Analysis :

The system evaluated three machine learning models for flood and risk prediction. The results showed that the Gradient Boosting model achieved the best overall performance across multiple evaluation metrics.

Model	Accuracy	Weighted F1	AUC
Random Forest	73.48%	0.7469	0.903
Gradient Boost	81.64%	0.8107	0.939
Logistic Regression	75.96%	0.7654	0.915



The Gradient Boosting model achieved the highest accuracy of 81.64% and the highest AUC score of 0.939, indicating strong predictive capability and a good balance between precision and recall across flood risk categories.

This model was therefore selected as the best-performing algorithm from the flood prediction system.

Feature Importance Analysis

It revealed that rainfall-related features played the most significant role in predicting flood risk. The most influential predictors included:

- Rainfall lag features from Albay station
- 24-hour rainfall accumulation
- Rainfall data from nearby stations such as Catanduanes

These results indicate that recent rainfall patterns and upstream precipitation are key indicators of flood risk.

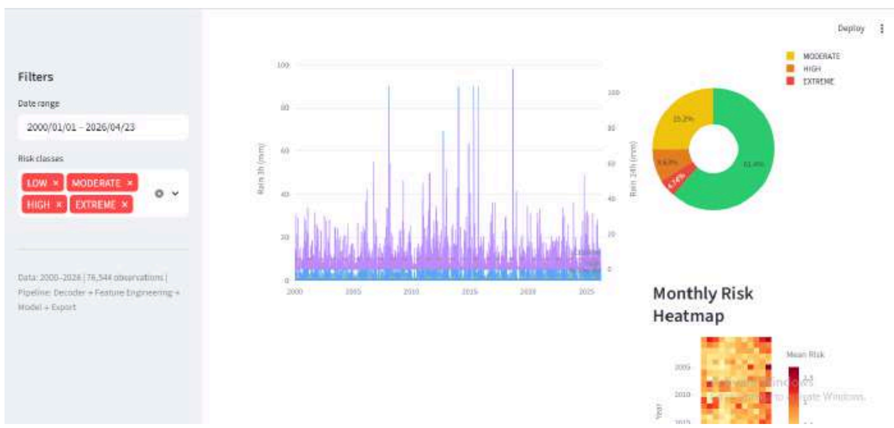
VISUALIZATION RESULTS

The system produced both interactive visualizations and static charts for analysis and reporting.

Interactive Dashboard Visualizations

Overview Tab

- Rainfall timeline showing precipitation levels over time
- Flood risk distribution pie chart

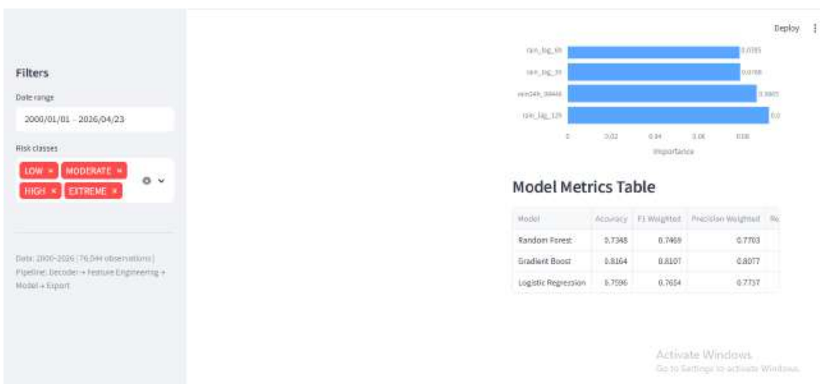


- Monthly risk heatmap
- Key performance indicator (KPI) cards showing high and extreme risk events

Model Performance Tab

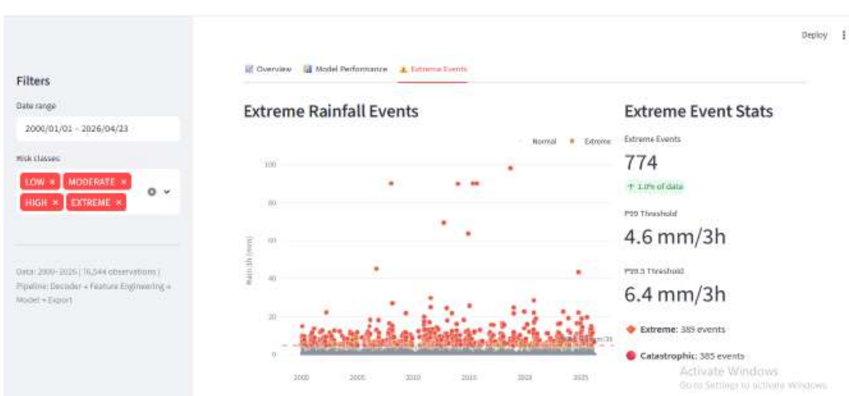
- Grouped bar chart comparing performance of all models
- Feature importance charts displaying top predictive variables

- Sortable metrics table for deeper analysis



Extreme Events Tab

- Scatter plot highlighting extreme rainfall events
- Statistical summaries of high-risk occurrences

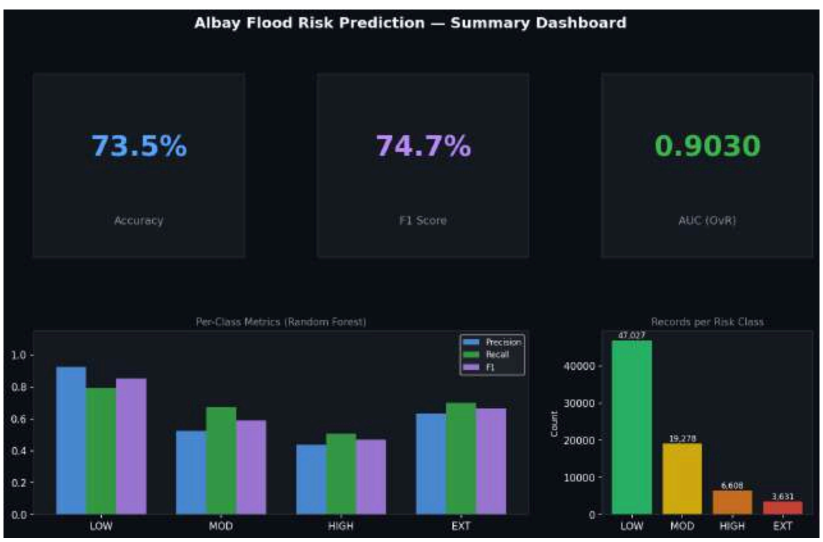


- Data tables listing significant events

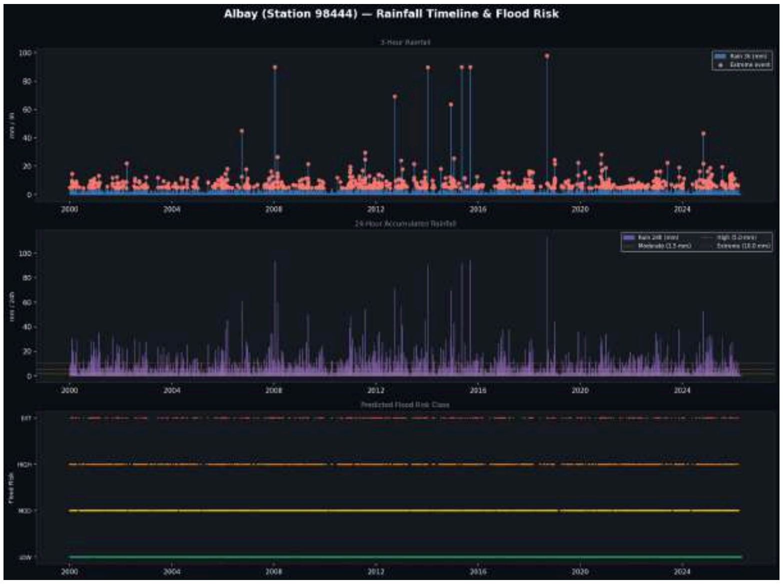
Static Output Visualization

The system also exports six static charts for documentation and reporting purposes :

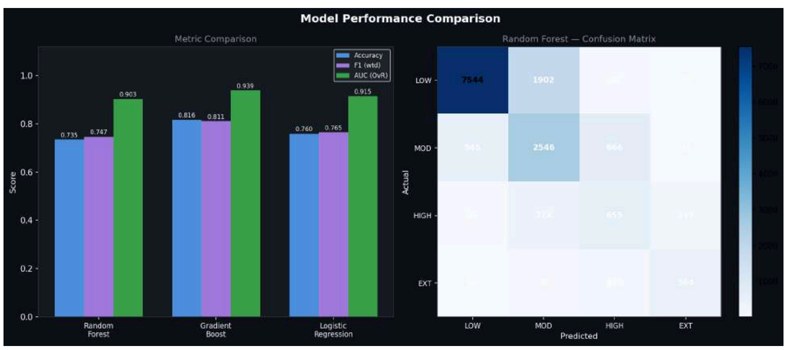
- Summary dashboard visualization



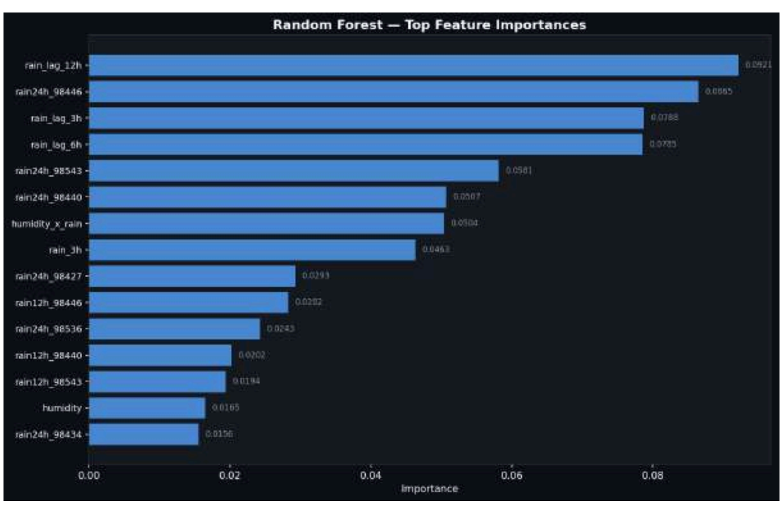
- Rainfall timeline chart



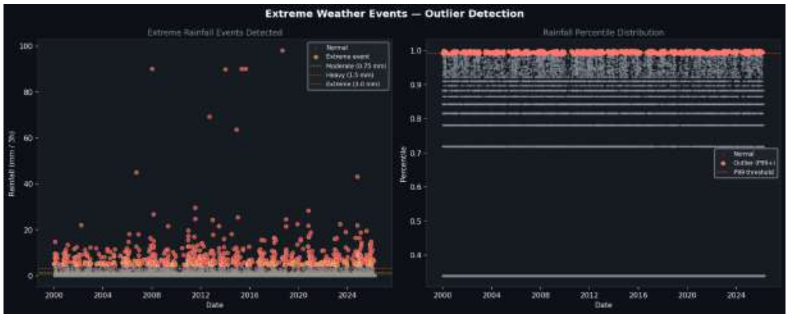
- Model comparison chart



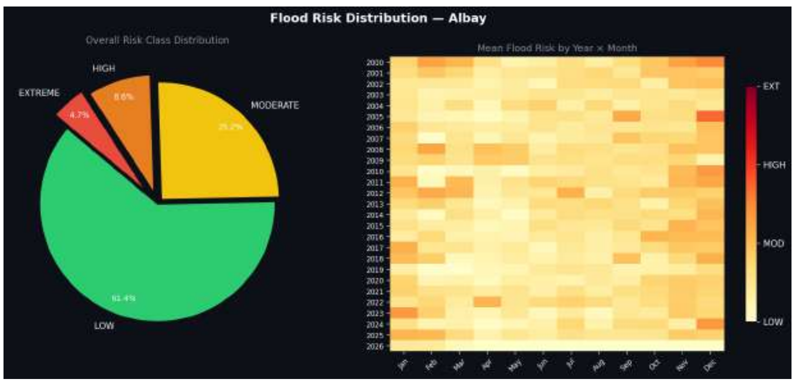
- Feature importance chart



- Extreme events scatter plot



- Risk distribution chart



These static charts provide a permanent record of the analysis and can be used in reports or presentations.

Challenges and Solutions :

During the implementation of the pipeline development and professional visualization tasks, the group encountered several technical and practical challenges. These challenges arose from the earlier methods and steps taken when accomplishing previous tasks. Below are the major ones identified along with the solutions applied:

Challenge 1: Target Leakage in Model Training

- Issue: Future rainfall features (e.g., rain_24h, rain_48h) were accidentally included in the feature set, causing unrealistically high model performance.
- Solution: Reviewed and removed all future-dependent features from the training data. Only past and current observations were used for prediction. Risk labels were generated strictly from the target 24-hour rainfall after feature engineering.

Challenge 2: Memory Overload with Large Datasets

- Issue: Loading all historical weather files into RAM at once caused memory spikes that risked out-of-memory crashes.
- Solution: Switched to a generator-based approach that processes station data sequentially with rolling transformations, reducing memory usage without losing temporal context.

Challenge 3: Model Export/Import Mismatches

- Issue: The pipeline was tied to a single model type, causing joblib deserialization failures when switching algorithms due to environment or object mismatches.
- Solution: Wrapped models in scikit-learn Pipeline objects that bundle preprocessors, classifiers, and required feature keys together for consistent serialization across training and deployment.

Challenge 4: Multi-Station Data Sync Gaps

- Issue: Stations going offline or logging asynchronously caused timestamp misalignments, and simple forward-fills introduced corrupted values into rolling features during storm peaks.
- Solution: Added distribution validation during interpolation and implemented mask matrices to flag missing windows, preventing bad data from affecting near-term predictions.

Challenge 5: Dashboard Lag on Re-renders

- Issue: Streamlit's top-to-bottom re-execution caused all charts to re-render on every user interaction, creating noticeable UI delays across multiple tabs.
- Solution: Isolated heavy Plotly charts inside tabs so they only render when selected, and pre-aggregated data before display to keep interactions responsive.

Challenge 6: Creating Professional and Interactive Visualizations

- Issue: Basic Matplotlib charts were not interactive or polished enough for stakeholder presentation.
- Solution: Integrated Plotly within Streamlit to create zoomable, hover-enabled charts. Implemented a clean three-tab layout with filters, dark theme, and conditional formatting for a professional look.

Conclusion :

The engineered Flood Risk Prediction System validates that marrying automated big data processing pipelines with modern ensemble learning models provides a highly reliable approach to regional hazard forecasting in Albay. By connecting raw meteorological observations to a optimized Gradient Boosting engine, the application accurately flags and evaluates developing flood risks.

The deployment of an interactive web application significantly lowers the barrier to entry for interpreting complex environmental data, providing clear insights for municipal planning and risk assessment. Future work will focus on integrating topological and soil-absorbency parameters, expanding the ingestion layer to include a wider array of telemetry stations, and setting up the core pipeline for automated, real-time alert broadcasts.